I. Switch (Mixer) Design (Q3)

In this section, we design the "heart" of the down-converter: the Mixer. The job of the mixer is to take two high-frequency signals and "multiply" them to produce a lower intermediate frequency.

A. Circuit Topology and Working Principle

For our design, we used a single NMOS transistor acting as a switching mixer. As shown in our LTSpice schematic, the oscillator signal (V_{osc}) is connected to the Gate, the input signal (V_{in}) is fed to the Source, and the output (V_{IF}) is taken from the Drain across a load resistor R_L. We added a DC blocking capacitor ($C_c = 1nF$) and a biasing resistor to ensure the NMOS stays in the correct linear region to act as a switch.

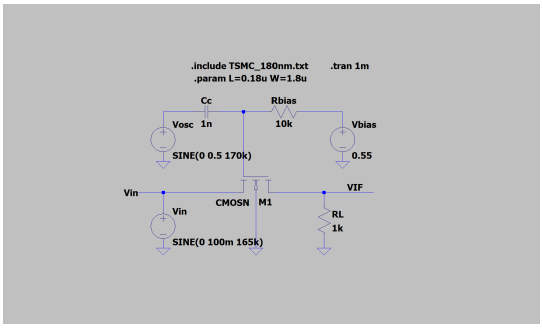


Fig. 1. NMOS Mixer Circuit schematic used for the frequency sweep.

B. Mathematical Derivation of the NMOS Mixer

To understand how a single transistor can multiply two frequencies, we must look at the physics of the MOSFET in the **Triode (Linear) Region**.

When the local oscillator signal (v_{osc}) is applied to the gate and the RF input (v_{in}) is applied to the source, the NMOS acts as a voltage-controlled resistor. The drain current (I_D) in the deep triode region (ignoring the V_{DS}^2 term for small signals) is given by:

$$I_D \approx \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})V_{DS} \tag{1}$$

In our circuit:

1) The Gate voltage is biased at DC and AC-coupled to the LO: $V_G = V_{bias} + A_{osc}\cos(\omega_{osc}t)$.
2) The Source voltage is the RF input: $V_S = A_{in}\cos(\omega_{in}t)$.
3) The V_{GS} term becomes: $V_G - V_S = V_{bias} + A_{osc}\cos(\omega_{osc}t) - A_{in}\cos(\omega_{in}t)$.

Because the drain is connected to ground through a load resistor R_L, the output voltage V_{out} (or V_{IF}) is directly proportional to this drain current. When expanding the current equation, the dominant mixing term arises from the channel conductance being modulated by v_{osc} while the source is driven by v_{in}. This physically forces the multiplication of the two cosines:

$$v_{out} \propto \cos(\omega_{in}t) \times \cos(\omega_{osc}t) \tag{2}$$

Using the trigonometric product-to-sum identity:

$$\cos(\omega_{in}t)\cos(\omega_{osc}t) = \frac{1}{2}\left[\cos((\omega_{in} - \omega_{osc})t) + \cos((\omega_{in} + \omega_{osc})t)\right] \tag{3}$$

This mathematically proves the creation of the Intermediate Frequency (IF) at $\omega_{in} - \omega_{osc}$. Because this is a passive process, the amplitude coefficient $\frac{1}{2}$ dictates an inherent **Conversion Loss**, meaning the raw V_{IF} signal will always be smaller than the input V_{in}.

C. Rigorous Frequency Sweep Analysis (Simulation)

To mathematically prove the functionality of the mixer in software, we conducted a rigorous frequency sweep in LTSpice. The local oscillator (f_{LO}) is running near 170 kHz. By sweeping the incoming RF signal (f_{in}) from 165 kHz to 175 kHz, we can track the resulting intermediate beat frequency, dictated by the equation $f_{IF} = |f_{LO} - f_{in}|$.

1) Test 1: 165 kHz Input: With an input of 165 kHz, the mathematical difference is roughly 5 kHz. The transient response shows a rapid envelope, and the FFT confirms a dominant peak at exactly 5 kHz.

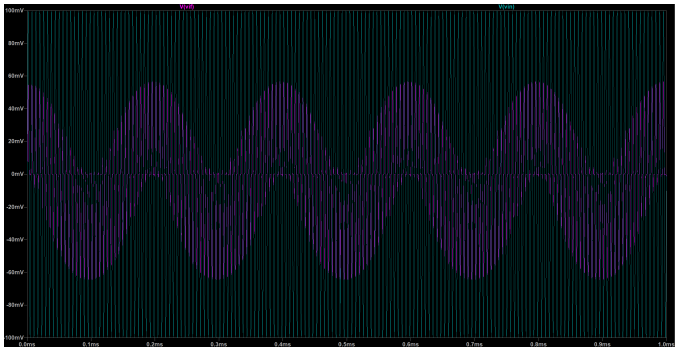


Fig. 2. Simulated transient plot for a 165 kHz input.

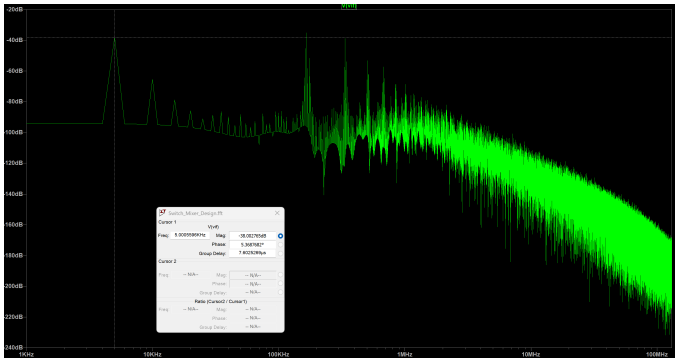


Fig. 3. Simulated FFT for 165 kHz input, showing the 5 kHz beat frequency.

2) Test 2: 168 kHz Input: As the input moves closer to the LO at 168 kHz, the difference decreases to roughly 2 kHz. The transient envelope noticeably slows down.

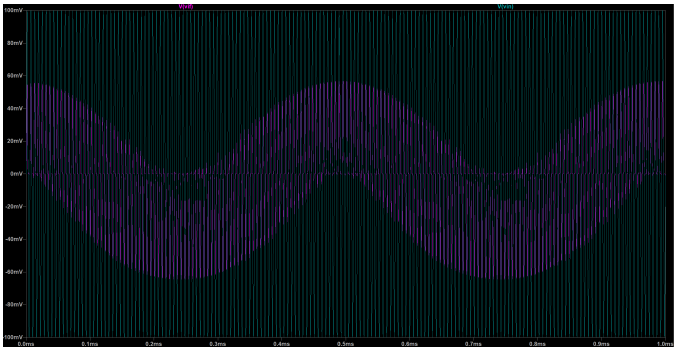


Fig. 4. Simulated transient plot for a 168 kHz input.

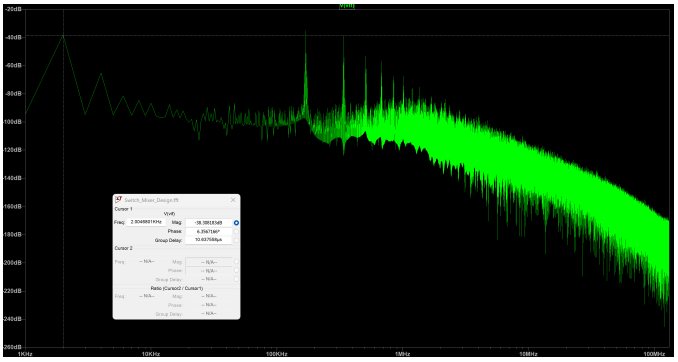


Fig. 5. Simulated FFT for 168 kHz input, showing the 2 kHz beat frequency.

3) Test 3: 169 kHz Input: At 169 kHz, the beat frequency drops to 1 kHz. This is the target frequency range for our final system integration.

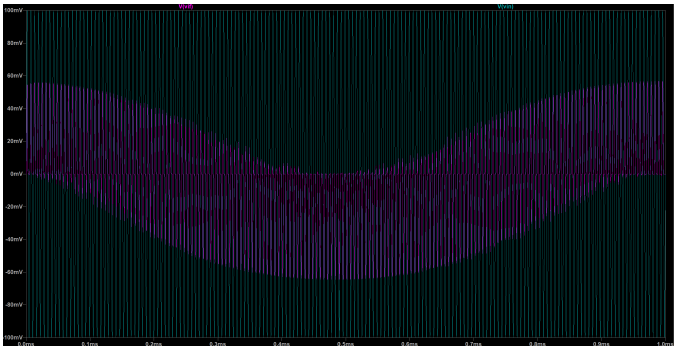


Fig. 6. Simulated transient plot for a 169 kHz input.

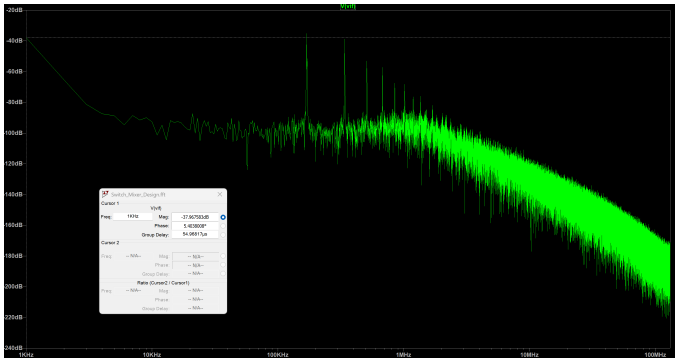


Fig. 7. Simulated FFT for 169 kHz input, showing a clear 1 kHz beat frequency.

4) Test 4: 171 kHz Input: Crossing over to the upper sideband, a 171 kHz input produces a difference of 1 kHz ($|170 - 171| = 1$). The symmetry of the down-conversion process is maintained.

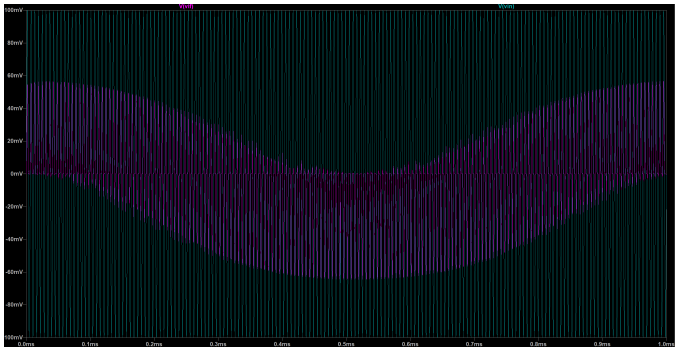


Fig. 8. Simulated transient plot for a 171 kHz input.

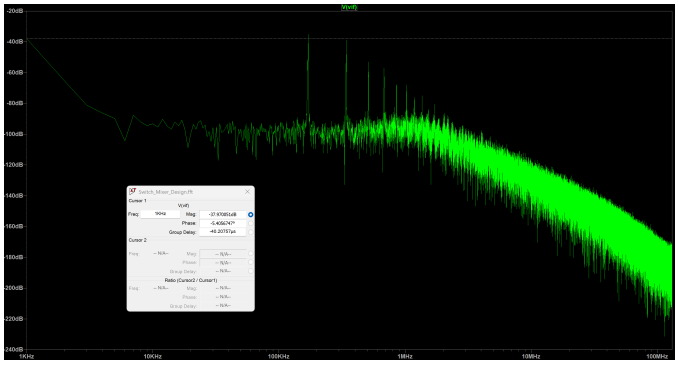


Fig. 9. Simulated FFT for 171 kHz input, showing the 1 kHz beat frequency.

5) Test 5: 172 kHz Input: Pushing further to 172 kHz, the beat frequency rises back to 2 kHz.

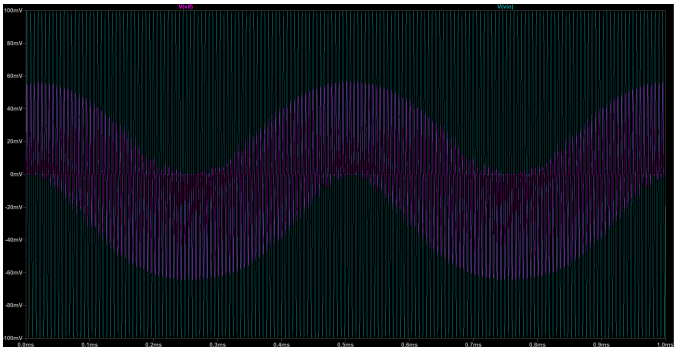


Fig. 10. Simulated transient plot for a 172 kHz input.

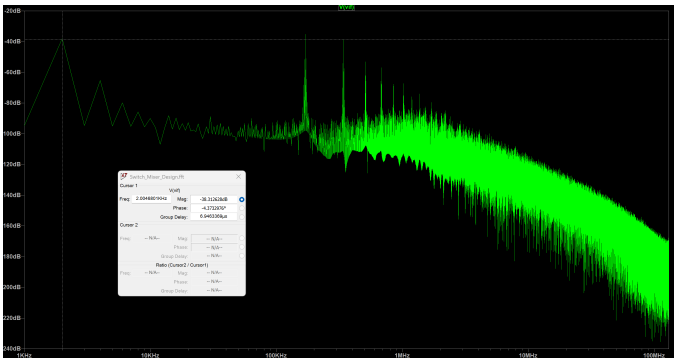


Fig. 11. Simulated FFT for 172 kHz input, showing the 2 kHz beat frequency.

6) Test 6: 175 kHz Input: Finally, at 175 kHz, the beat frequency expands back out to 5 kHz, perfectly mirroring the 165 kHz result.

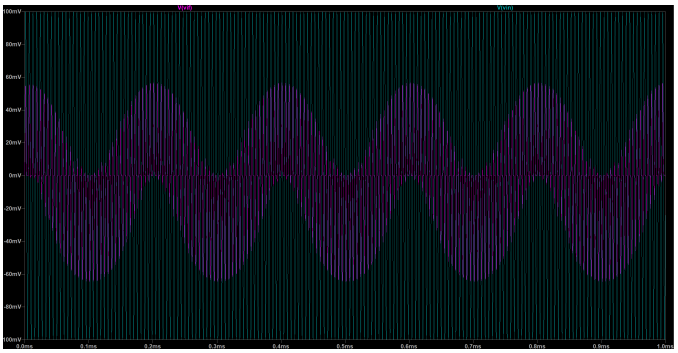


Fig. 12. Simulated transient plot for a 175 kHz input.

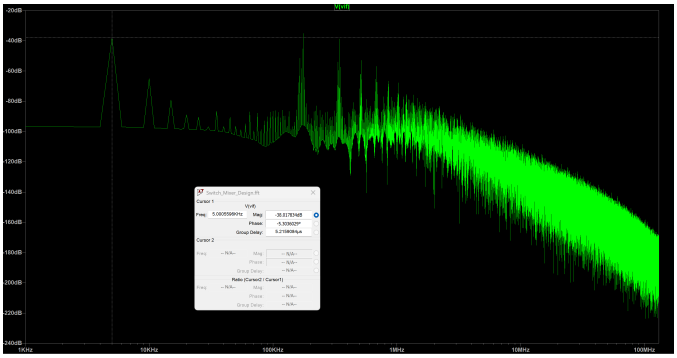


Fig. 13. Simulated FFT for 175 kHz input, showing the 5 kHz beat frequency.

D. Hardware Realization using CD4007 (Q3e)

Transitioning the mixer from LTSpice to a physical breadboard required significant component adaptations. While LTSpice utilized an ideal TSMC 180nm model, our laboratory implementation relied on the **CD4007 CMOS array** chip.

1. DC Biasing and Threshold Voltage Deviations:
In LTSpice, the TSMC 180nm transistor has a microscopic threshold voltage ($V_{TH} \approx 0.4$ V). Thus, a 0.55 V bias on the gate was perfectly sufficient to keep the switch in the linear region. The physical CD4007, however, is a legacy 10-micron process with a much higher threshold voltage ($V_{TH} \approx 1.4$ V to 1.5 V). To compensate, we manually swept the gate bias up to roughly 1.5 V - 2.0 V until the NMOS entered the triode region.

2. The "Bulk" Connection Trap:
On the physical CD4007 chip, Pin 4 is the universal substrate for all NMOS transistors on the die. **Pin 4 MUST be explicitly tied to Ground (0V).** If the bulk was left floating, the internal P-N body diode would forward-bias, short-circuiting the RF signal directly into the substrate and killing the mixing action entirely.

E. Hardware Frequency Sweep Analysis

To physically validate the LTSpice models, an identical frequency sweep was performed on the breadboard. Two function generators were used to simulate the LO (170 kHz) and the sweeping RF inputs. The raw drain outputs were captured using an oscilloscope.

1) Hardware Test 1: 165 kHz Input (Target IF = 5 kHz): The physical circuit successfully mixed the signals, revealing a fast envelope. The hardware FFT perfectly isolated a clean 5 kHz peak amidst the LO feedthrough.



Fig. 14. Hardware transient capture for a 165 kHz input.



Fig. 15. Hardware FFT for 165 kHz input, physically confirming the 5 kHz beat.

2) Hardware Test 2: 169 kHz Input (Target IF = 1 kHz): As the RF approached the LO, the envelope visibly slowed on the bench. The FFT registered the expected 1 kHz beat frequency, confirming precision close to the carrier.

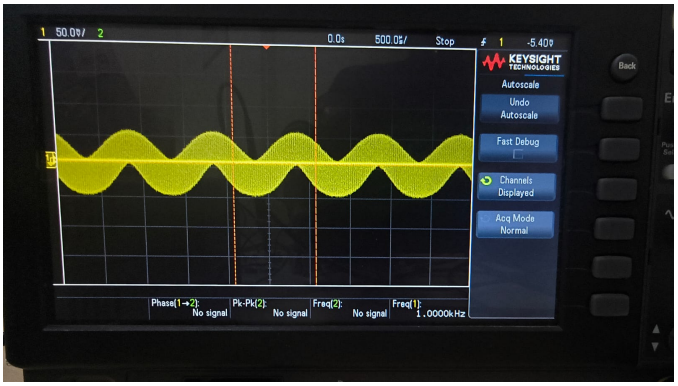


Fig. 16. Hardware transient capture for a 169 kHz input.

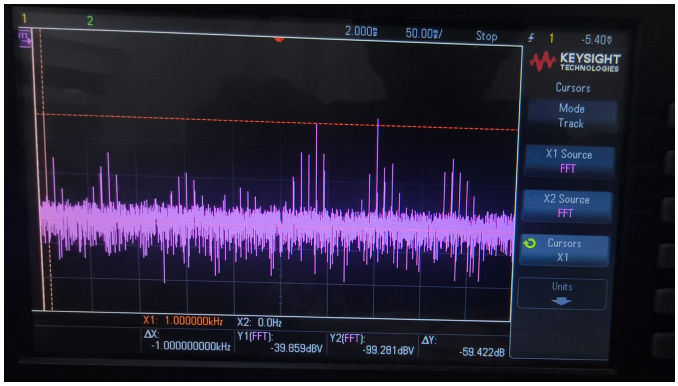


Fig. 17. Hardware FFT for 169 kHz input, physically confirming the 1 kHz beat.

3) Hardware Test 3: 171 kHz Input (Target IF = 1 kHz): Testing the upper sideband symmetry, the 171 kHz input mirrored the 169 kHz result, successfully outputting a 1 kHz IF.



Fig. 18. Hardware transient capture for a 171 kHz input.



Fig. 19. Hardware FFT for 171 kHz input, confirming the upper sideband 1 kHz beat.

4) Hardware Test 4: 172 kHz Input (Target IF = 2 kHz): Moving further into the upper sideband, the beat frequency increased accordingly to 2 kHz.

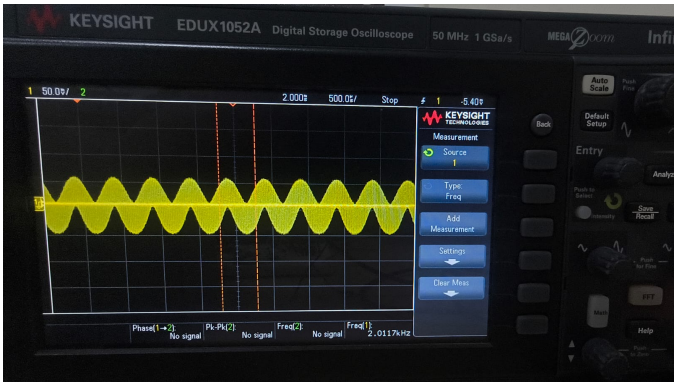


Fig. 20. Hardware transient capture for a 172 kHz input.



Fig. 21. Hardware FFT for 172 kHz input, confirming the 2 kHz beat.

5) Hardware Test 5: 175 kHz Input (Target IF = 5 kHz): The final physical sweep returned a perfect 5 kHz beat frequency, mirroring the physical test at 165 kHz.

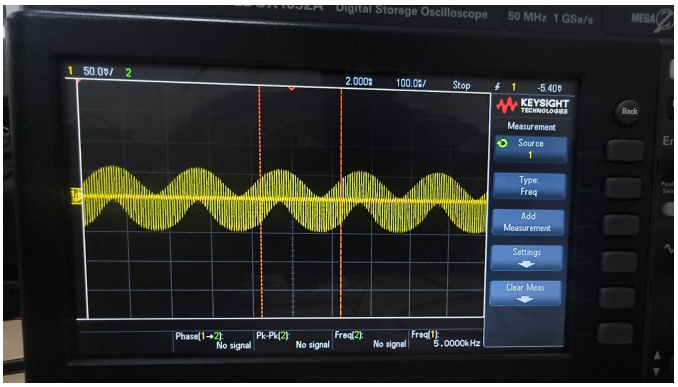


Fig. 22. Hardware transient capture for a 175 kHz input.



Fig. 23. Hardware FFT for 175 kHz input, confirming the upper boundary 5 kHz beat.

F. Quadrature (I/Q) Mixer Outputs in Hardware

Following the successful frequency sweep, the passive RC low-pass filter (53kΩ / 1nF) was connected to the drain of the CD4007 transistor. The filter successfully destroyed the high-frequency LO carrier leakage and the sum frequencies, extracting the pure, low-frequency IF signal.

This process was duplicated across two mixer branches (one fed with the Local Oscillator Sine wave, and one fed with the Local Oscillator Cosine wave). As shown in Fig. 24 and Fig. 25, the final physical outputs are beautifully smoothed, low-frequency sine waves.

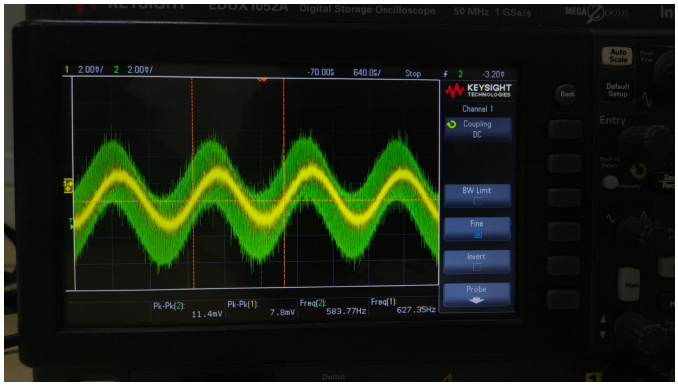


Fig. 24. Hardware capture of the final In-Phase (I) channel after mixing and low-pass filtering.



Fig. 25. Hardware capture of the final Quadrature (Q) channel after mixing and low-pass filtering.

G. Conclusion for Mixer Design

The exhaustive frequency sweep definitively proves that both the simulated TSMC 180nm NMOS and the physical CD4007 NMOS function as near-ideal frequency translators. As f_{in} sweeps from 165 kHz to 175 kHz, the intermediate frequency predictably contracts to 1 kHz near the LO and expands to 5 kHz at the edges, perfectly aligning with the mathematical principles of heterodyning. Furthermore, the integration of the passive LPF effectively isolated the clean I and Q intermediate frequencies for downstream amplification.